William Chenyin

Software Engineer Intern | Full-Stack Development | AI Applications | Test Automation

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Education

University of Waterloo — Waterloo, ON

Bachelor of Computer Science (Honours), Co-operative Program — *Sep. 2025 – Apr. 2030*

- GPA: **3.9/4.0**; President's Scholarship with Distinction; BC Provincial District/Authority Scholarship.
- Relevant coursework: Algorithm Design, Functional Programming, Linear Algebra, and Calculus.

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, C, C++, C#, Rust, SQL, Racket, HTML/CSS, Bash
Frameworks and Libraries: React, Next.js, Blazor, .NET, Vite, Tailwind CSS, ShadCN UI, NextAuth, Zod
AI Application Development: WebLLM, WebGPU, Qwen2.5, on-device inference, prompt specifications.
Testing and Automation: Playwright, WebdriverIO, Appium, BrowserStack, end-to-end testing, mobile testing, CI/CD
Data, APIs, and Tools: PostgreSQL, MongoDB, REST APIs, Git, Azure DevOps, Linux, Vercel, ANSYS HFSS

Professional Experience

Software QA Automation Engineer, Co-op — May 2026 – Aug. 2026

Method Integration (Method CRM) — *Toronto, ON*

- Led a **Playwright reliability refactor** across three end-to-end test suites, replacing fixed waits and fragile locators with state-based assertions and reusable utilities across critical CRM workflows.
- Documented test architecture, locator, assertion, and review standards adopted across an **11-person QA organization**, improving consistency in Playwright and WebdriverIO contributions.
- Built a mobile automation foundation covering **20+ iOS and Android flows** with TypeScript, WebdriverIO, Appium, and BrowserStack, then integrated execution into the team's daily CI/CD pipeline.
- Implemented backend validation for priority requirements in `ms-scheduler-api`, converting expected API behaviour and failure cases into executable service checks staged for production.
- Authored repository-level AI coding specifications (`CLAUDE.md` and `codex.md`) covering test structure, naming, locators, assertions, and review conventions for AI-assisted test generation.

Math Tutor — Jun. 2023 – Aug. 2023

RootMaths Math Tutoring — *Coquitlam, BC*

- Taught number theory, combinatorics, and geometry to students aged 8 to 10 through more than **50 worked problems**.
- Converted abstract contest-math concepts into repeatable problem-solving strategies, helping students explain their reasoning.

Projects

GoodLife.AI | Live App | Source | *React, TypeScript, WebLLM, WebGPU* — Aug. 2026

- Built a local-first life-coaching PWA that converts questionnaire responses into three deterministic recommendations and a personalized seven-day habit plan spanning money, sleep, health, relationships, and home.
- Integrated a quantized **500M-parameter Qwen2.5 small language model** with WebLLM and WebGPU, enabling private, on-device conversational inference without an API key or server-side model calls.
- Implemented domain classification and deterministic fallbacks for financial, health, housing, and crisis prompts; persisted profiles, streaks, progress, and chat locally with portable JSON export.

HardHaq '25 | Source | Presentation | *Python, ANSYS HFSS* — Nov. 2025

- Won **first place** among **20+ teams** at Canada's only quantum hardware hackathon, then presented the completed design and simulation results to expert judges.
- Automated ANSYS HFSS parameter sweeps, project setup, and repeated simulation tasks with Python, increasing experiment throughput by approximately **25x** during the hackathon.
- Designed and simulated a superconducting qubit with **35.8 MHz coupling** against **900 MHz detuning**, iterating on device geometry and electromagnetic behaviour.

StockStalker | Live Dashboard | Source | *React, TypeScript, Financial APIs* — Jun. 2026

- Built a responsive stock and ETF research dashboard with ticker search, historical price charts, market news, and news-based Quick Take summaries backed by cited articles.
- Implemented ETF comparison and portfolio analytics across expense ratio, AUM, yield, volatility, holdings, unrealized P/L, allocation, and automatically generated portfolio risk flags.

The Air Outside | Play | Source | *Blazor WebAssembly, .NET, C#, Ink* — May 2026

- Built a client-side visual novel with **15 scenes** and **9 endings**, using an Ink DSL narrative layer compiled into an intermediate JSON representation for a C#/Blazor renderer.
- Created more than **100 SVG illustrations** and a procedural Web Audio synthesizer without external media assets, then deployed to Vercel and packaged the project as a Windows desktop application.